University of the Philippines - Baguio

Archie Junio's Reference Notebook

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1. Data Structures

1.1. Segment Tree.

```
class SegmentTree:
    def __init__(self, data):
        self.n = len(data)
        self.tree = [0] * (2 * self.n)
        for i in range(self.n):
            self.tree[self.n + i] = data[i]
        for i in range(self.n - 1, 0, -1):
            self.tree[i] = self.tree[i << 1] + self.tree[i << 1 | 1]

    def update(self, p, value):
        p += self.n
        self.tree[p] = value
        while p > 1:
            self.tree[p >> 1] = self.tree[p] + self.tree[p ^ 1]
            p >>= 1

    def query(self, l, r):
        res = 0
        l += self.n
        r += self.n
        while l < r:
            if l & 1:
                res += self.tree[l]
                l += 1
            if r & 1:
                r -= 1
                res += self.tree[r]
            l >>= 1
            r >>= 1
        return res
```

1.2. Union Find.

```
class DSU:
    def __init__(self, n):
        self.parent = list(range(n))
        self.size = [1] * n

    def find(self, i):
        if self.parent[i] == i:
            return i
        self.parent[i] = self.find(self.parent[i])
        return self.parent[i]

    def union(self, i, j):
        root_i = self.find(i)
        root_j = self.find(j)

        if root_i != root_j:
            if self.size[root_i] < self.size[root_j]:
                root_i, root_j = root_j, root_i
            self.parent[root_j] = root_i
            self.size[root_i] += self.size[root_j]
            return True
        return False
```

```
--- def get_size(self, i):
------- return self.size[self.find(i)]
```

1.3. 2D Sparse Table.

```
import math

class SparseTable2D:
--- def __init__(self, A):
------- self.A = A
------- self.n = len(A)
------- self.m = len(A[0])

------- self.LOGN = int(math.log2(self.n)) + 1
------- self.LOGM = int(math.log2(self.m)) + 1

------- self.lg = [0] * (max(self.n, self.m) + 1)
------- for i in range(2, len(self.lg)):
----------- self.lg[i] = self.lg[i // 2] + 1

------- self.st = [[[[0] * self.m for _ in range(self.n)]
------------------- for _ in range(self.LOGM)]
------------------ for _ in range(self.LOGN)]

------- self.build()

--- def build(self):
------- for i in range(self.n):
----------- for j in range(self.m):
--------------- self.st[0][0][i][j] = self.A[i][j]

------- for bj in range(1, self.LOGM):
----------- for i in range(self.n):
--------------- for j in range(self.m - (1 << bj) + 1):
------------------- self.st[0][bj][i][j] = max(
----------------------- self.st[0][bj - 1][i][j],
----------------------- self.st[0][bj - 1][i][j + (1 << (bj - 1))]
------------------- )

------- for bi in range(1, self.LOGN):
----------- for i in range(self.n - (1 << bi) + 1):
--------------- for j in range(self.m):
------------------- self.st[bi][0][i][j] = max(
----------------------- self.st[bi - 1][0][i][j],
----------------------- self.st[bi - 1][0][i + (1 << (bi - 1))][j]
------------------- )

------- for bi in range(1, self.LOGN):
----------- for bj in range(1, self.LOGM):
--------------- for i in range(self.n - (1 << bi) + 1):
------------------- for j in range(self.m - (1 << bj) + 1):
----------------------- self.st[bi][bj][i][j] = max(
--------------------------- self.st[bi - 1][bj - 1][i][j],
--------------------------- self.st[bi - 1][bj - 1][i + (1 << (bi - 1))][j],
--------------------------- self.st[bi - 1][bj - 1][i][j + (1 << (bj - 1))],
--------------------------- self.st[bi - 1][bj - 1][i + (1 << (bi - 1))][j + (1 << (bj - 1))]
----------------------- )

--- def query(self, x1, y1, x2, y2):
------- kx = self.lg[x2 - x1 + 1]
------- ky = self.lg[y2 - y1 + 1]

------- return max(
----------- self.st[kx][ky][x1][y1],
----------- self.st[kx][ky][x2 - (1 << kx) + 1][y1],
----------- self.st[kx][ky][x1][y2 - (1 << ky) + 1],
----------- self.st[kx][ky][x2 - (1 << kx) + 1][y2 - (1 << ky) + 1]
------- )
# Input Format: st = SparseTable2D(grid)
#               print(st.query(1, 1, 2, 2))
```

2. Dynamic Programming

2.1. Lichao Tree.

```
class Line:
--- __slots__ = ['m', 'c']
--- def __init__(self, m=0, c=float('-inf')):
------- self.m = m
------- self.c = c

--- def eval(self, x):
------- return self.m * x + self.c

class Node:
--- __slots__ = ['line', 'left', 'right']
--- def __init__(self):
------- self.line = Line()
------- self.left = None
------- self.right = None

class LiChaoTree:
--- def __init__(self, min_x=-10**12, max_x=10**12):
------- self.root = Node()
------- self.min_x = min_x
------- self.max_x = max_x

--- def add_line(self, m, c):
------- self._insert(self.root, self.min_x, self.max_x, Line(m, c))

--- def _insert(self, node, l, r, new_line):
------- mid = (l + r) // 2

------- left_better = new_line.eval(l) > node.line.eval(l)
------- mid_better = new_line.eval(mid) > node.line.eval(mid)

------- if mid_better:
----------- node.line, new_line = new_line, node.line
------- if r - l <= 1:
----------- return

------- if left_better != mid_better:
----------- if not node.left:
--------------- node.left = Node()
----------- self._insert(node.left, l, mid, new_line)
------- else:
----------- if not node.right:
--------------- node.right = Node()
----------- self._insert(node.right, mid, r, new_line)

--- def query(self, x):
------- return self._query(self.root, self.min_x, self.max_x, x)

--- def _query(self, node, l, r, x):
------- if not node:
----------- return float('-inf')
------- if r - l <= 1:
----------- return node.line.eval(x)

------- mid = (l + r) // 2
------- res = node.line.eval(x)

------- if x < mid:
----------- res = max(res, self._query(node.left, l, mid, x))
------- else:
----------- res = max(res, self._query(node.right, mid, r, x))

------- return res
```

2.2. Divide And Conquer.

```
INF = float('inf')

def solve_dp(g, k_L, k_R, n_L, n_R, dp, cost_func):
--- if n_L > n_R:
------- return

--- n_M = (n_L + n_R) // 2
--- best_k = -1
--- dp[g][n_M] = INF

--- for k in range(k_L, min(n_M, k_R) + 1):
------- current_val = dp[g - 1][k] + cost_func(k + 1, n_M)
------- if current_val < dp[g][n_M]:
----------- dp[g][n_M] = current_val
----------- best_k = k

--- solve_dp(g, k_L, best_k, n_L, n_M - 1, dp, cost_func)
--- solve_dp(g, best_k, k_R, n_M + 1, n_R, dp, cost_func)
```

3. Geometry

3.1. Angles And Rotations.

```
def angle(a, b, c): # angle formed by abc in radians: PI < x <= PI
--- return abs((arg(a-b) - arg(c-b)) % (2 * math.pi))
def rotate(p, a, d): # rotate point a about pivot p CCW at d radians
--- return p + (a - p) * complex(math.cos(d), math.sin(d))
```

3.2. Distances.

```
# Distance: Point to 2D Line (ax + by + c = 0)
def dist_pt_line_eq(p, a, b, c):
--- return abs(a*p.real + b*p.imag + c) / math.hypot(a, b)

# Distance: Point to 2D Line through points a and b
def dist_pt_line(p, a, b):
--- cp = (p - a).conjugate() * (b - a)
--- return abs(cp.imag) / abs(b - a)

# Distance: Point to 3D Plane (ax + by + cz + d = 0)
def dist_pt_plane(p, a, b, c, d):
--- # math.hypot supports 3+ arguments in Python 3.8+
--- return abs(a*p.x + b*p.y + c*p.z + d) / math.hypot(a,b,c)

# Distance: Shortest distance between 3D lines AB and CD
def dist_line_3d(A, B, C, D):
--- u, v, w = B - A, D - C, A - C
--- a, b = u.dot(u), u.dot(v)
--- c, d, e = v.dot(v), u.dot(w), v.dot(w)
--- det = a * c - b * b
--- if det < 1e-9: # Parallel lines
------- s, t = 0.0, d / b if b > c else e / c
--- else: # Skew lines
------- s = (b * e - c * d) / det
------- t = (a * e - b * d) / det
--- p1 = P3(A.x + u.x*s, A.y + u.y*s, A.z + u.z*s)
--- p2 = P3(C.x + v.x*t, C.y + v.y*t, C.z + v.z*t)
--- vec = p1 - p2
--- return math.sqrt(vec.dot(vec))
```

3.3. Intersections.

3.3.1. *Circle Line Intersection.*

```
EPS = 1e-9
def projLine(p, a, b): # Project point p onto line segment ab
--- v = b - a
--- scalar = ((p - a) * v.conjugate()).real / abs(v)**2
--- return a + v * scalar

def rotate(c, p, t): # Rotate point p around center c by t radians
```

```
    return c + (p - c) * cmath.rect(1, t)

def CL_inter(c, r, a, b):
    p = projLine(c, a, b)
    d = abs(c - p)
    ans = []

    if d > r + EPS: pass # None
    elif d > r - EPS: ans.append(p) # Tangent
    elif d < EPS: # Diameter
        v = r * (b - a) / abs(b - a)
        ans.append(c + v)
        ans.append(c - v)
    else: # Secant
        t = math.acos(d / r)
        p = c + (p - c) * r / d
        ans.append(rotate(c, p, t))
        ans.append(rotate(c, p, -t))
    return ans
```

3.3.2. *Circle Circle Intersection.*

```
EPS = 1e-9
def rotate(c, p, t):
    # Rotate point p around center c by t radians
    return c + (p - c) * cmath.rect(1, t)

def CC_inter(c1, r1, c2, r2):
    d = abs(c1 - c2)
    ans = []
    if d < EPS: # Concentric circles
        if abs(r1 - r2) < EPS:
            pass # Infinite intersections (same circle)
    elif r1 < EPS: # Circle 1 is a point
        if abs(d - r2) < EPS:
            ans.append(c1)
    else: # General case using Law of Cosines
        s = (r1*r1 + d*d - r2*r2) / (2*r1*d)
        s = max(-1.0, min(1.0, s)) # Prevent float domain errors
        t = math.acos(s)
        # Point on C1's radius pointing towards C2
        mid = c1 + (c2 - c1) * r1 / d
        ans.append(rotate(c1, mid, t))
        if abs(math.sin(t)) >= EPS:
            # FIXED: Rotate around c1, not c2!
            ans.append(rotate(c1, mid, -t))
    return ans
```

3.3.3. *Line Intersect.*

```
def line_inter_fast(a, b, c, d):
    A = cross(d - a, b - a)
    B = cross(c - a, b - a)
    return (B * d - A * c) / (B - A) if B - A else None
```

3.4. Closest Pair.

```
def closest_pair(pts):
    if len(pts) <= 1: return float('inf')
    px = sorted(pts) # Automatically sorts by x, then y
    def solve(l, r): # Base case: Brute force for small partitions
        if r - l <= 3:
            d = float('inf')
            for i in range(l, r):
                for j in range(i + 1, r):
                    d = min(d, math.dist(px[i], px[j]))
            return d
        mid = (l + r) // 2
        mid_x = px[mid][0]
        # Recursively find the minimum in both halves
        d = min(solve(l, mid), solve(mid, r))
        # Gather points in the middle strip
        strip = [p for p in px[l:r] if abs(p[0] - mid_x) < d]
        strip.sort(key=lambda p: p[1]) # Sort strip by y
        # Check at most 7 geometric neighbors in the strip
        for i, p1 in enumerate(strip):
            for p2 in strip[i+1 : i+8]:
                d = min(d, math.dist(p1, p2))
        return d
    return solve(0, len(px))
```

3.5. Convex Hull.

3.5.1. *Convex Hull.*

```
def cross(o, a, b): # 2D cross product of OA and OB vectors
    return (a[0]-o[0])*(b[1]-o[1]) - (a[1]-o[1])*(b[0]-o[0])

def convex_hull(pts): # Sorts by x, then by y automatically
    pts = sorted(set(pts))
    if len(pts) <= 1:
        return pts
    def build(iterable):
        h = []
        for p in iterable:
            while len(h) >= 2 and cross(h[-2], h[-1], p) <= 0:
                h.pop()
            h.append(p)
        return h
    lower = build(pts)
    upper = build(reversed(pts))
    return lower[:-1] + upper[:-1]
```

3.5.2. *Convex Hull 3D.*

```
class P3:
    __slots__ = ['x', 'y', 'z']
    def __init__(self, x=0, y=0, z=0):
        self.x, self.y, self.z = x, y, z
    def __sub__(self, o):
        return P3(self.x-o.x, self.y-o.y, self.z-o.z)
    def cross(self, o):
        return P3(self.y*o.z - self.z*o.y,
                  self.z*o.x - self.x*o.z,
                  self.x*o.y - self.y*o.x)
    def dot(self, o):
        return self.x*o.x + self.y*o.y + self.z*o.z
    def __eq__(self, o):
        return (self.x, self.y, self.z) == (o.x, o.y, o.z)
    def __lt__(self, o):
        return (self.x, self.y, self.z) < (o.x, o.y, o.z)

class Face:
    __slots__ = ['p', 'q']
    def __init__(self, p, q):
        self.p = p # Tuple of 3 point indices
        self.q = q # Normal vector (P3 object)

def hull_3D(pts):
    n = len(pts)
    faces = []
    dead = [[True] * n for _ in range(n)]
    def add_face(a, b, c):
        q = (pts[b] - pts[a]).cross(pts[c] - pts[a])
        faces.append(Face((a, b, c), q))
        dead[a][b] = dead[b][c] = dead[c][a] = False
    # Initialize with a degenerate tetrahedron
    add_face(0, 1, 2)
    add_face(0, 2, 1)
    for i in range(3, n):
        f_inv = []
        for f in faces:
            # If face is visible from the point
            if (pts[i] - pts[f.p[0]]).dot(f.q) > 0:
                for j in range(3):
                    dead[f.p[j]][f.p[(j+1)%3]] = True
            else:
                f_inv.append(f)
        faces = []
        for f in f_inv:
            for j in range(3):
                a, b = f.p[j], f.p[(j+1)%3]
                # If reverse edge is dead but forward is alive
                if dead[b][a]:
                    add_face(b, a, i)
        faces.extend(f_inv)
    return faces
```

3.6. Areas.

3.6.1. *Triangle Area.*

```
def area(a, b, c):
    s = (a + b + c) / 2
    return (s*(s-a)*(s-b)*(s-c)) ** 0.5
```

3.6.2. *Polygon Area.*

```
def area(p, n):
    a = 0
    for i in range(n):
        j = (i - 1) % n
        a += cross(p[i], p[j])
    return abs(a) / 2
```

3.7. Dot Cross.

```
def dot(a, b):
    return a.x * b.x + a.y * b.y # + a.z * b.z
def cross(a, b):
    return a.x * b.y - a.y * b.x
def cross(a, b, c):
    return cross(a, b) + cross(b, c) + cross(c, a)
def cross3D(a, b):
    return point3(a.x*b.y - a.y*b.x, a.y*b.z - a.z*b.y, a.z*b.x -
↪   a.x*b.z)
```

3.8. Line Sweep Closest Pair.

```
import math

def dist_sq(p1, p2):
    return (p1[0] - p2[0])**2 + (p1[1] - p2[1])**2

def closest_pair(points):
    points.sort()

    def solve(pts):
        n = len(pts)

        if n <= 3:
            return min(dist_sq(pts[i], pts[j])
                       for i in range(n)
                       for j in range(i + 1, n)) if n > 1 else
↪   float('inf')

        mid = n // 2
        mid_x = pts[mid][0]

        left = pts[:mid]
        right = pts[mid:]

        d = min(solve(left), solve(right))

        strip = [p for p in pts if (p[0] - mid_x) ** 2 < d]
        strip.sort(key=lambda p: p[1])
```

```
        for i in range(len(strip)):
            for j in range(i + 1, min(i + 7, len(strip))):
                d = min(d, dist_sq(strip[i], strip[j]))

    return d

return solve(points)
```

4. Graphs

4.1. Max Flow.

4.1.1. *Dinic.*

```
INF = float('inf')
class Dinic:
    def __init__(self, n, s, t):
        self.n, self.s, self.t = n, s, t
        self.edges = []
        self.adj = [[] for _ in range(n)]

    def add_edge(self, u, v, c, bi=False):
        self.adj[u].append(len(self.edges))
        self.edges.append([v, c, 0])
        self.adj[v].append(len(self.edges))
        self.edges.append([u, c if bi else 0, 0])

    def bfs(self):
        self.dist = [-1] * self.n
        self.dist[self.s] = 0
        q = deque([self.s])
        while q:
            u = q.popleft()
            for i in self.adj[u]:
                v, c, f = self.edges[i]
                if self.dist[v] < 0 and c > f:
                    self.dist[v] = self.dist[u] + 1
                    q.append(v)
        return self.dist[self.t] != -1

    def dfs(self, u, pushed):
        if pushed == 0 or u == self.t: return pushed
        while self.ptr[u] < len(self.adj[u]):
            i = self.adj[u][self.ptr[u]]
            v, c, f = self.edges[i]
            if self.dist[u]+1 == self.dist[v] and c > f:
                tr = self.dfs(v, min(pushed, c - f))
                if tr > 0:
                    self.edges[i][2] += tr
                    self.edges[i^1][2] -= tr
                    return tr
            self.ptr[u] += 1
        return 0

    def calc_max_flow(self):
        flow = 0
        while self.bfs():
            self.ptr = [0] * self.n
            while True:
                pushed = self.dfs(self.s, INF)
                if not pushed:
                    break
                flow += pushed
        return flow
```

4.1.2. *Edmonds Karp.*

```
INF = float('inf')
class EdmondsKarp:
    def __init__(self, n, s, t):
        self.n, self.s, self.t = n, s, t
        self.adj = [[] for _ in range(n)]
        self.res = [[0] * n for _ in range(n)]

    def add_edge(self, u, v, w):
        self.adj[u].append(v)
        self.adj[v].append(u)
        self.res[u][v] += w

    def bfs(self):
        self.par = [-1] * self.n
        self.par[self.s] = self.s
        q = deque([self.s])
        while q:
            u = q.popleft()
            for v in self.adj[u]:
                if self.res[u][v] > 0 and self.par[v] == -1:
                    self.par[v] = u
                    if v == self.t:
                        return True
                    q.append(v)
        return False

    def calc_max_flow(self):
        ans = 0
        while self.bfs():
            flow = INF
            u = self.t
            while u != self.s:
                p = self.par[u]
                flow = min(flow, self.res[p][u])
                u = p
            u = self.t
            while u != self.s:
                p = self.par[u]
                self.res[p][u] -= flow
                self.res[u][p] += flow
                u = p
            ans += flow
        return ans
```

4.1.3. *Bipartite Matching.*

```
from collections import deque

INF = float('inf')
class BipartiteGraph:
    def __init__(self, n, m):
        self.n = n
        self.m = m
        self.adj = [[] for _ in range(n)]
        self.L = [-1] * n
        self.R = [-1] * m
        self.dist = [INF] * (n + 1)

    def add_edge(self, u, v):
        self.adj[u].append(v)

    def bfs(self):
        queue = deque()
        for v in range(self.n):
            if self.L[v] == -1:
                self.dist[v] = 0
                queue.append(v)
            else:
                self.dist[v] = INF

        self.dist[self.n] = INF

        while queue:
            v = queue.popleft()
            if self.dist[v] < self.dist[self.n]:
                for u in self.adj[v]:
                    next_v = self.R[u]
                    dist_next = self.dist[next_v] if next_v != -1 else
↪   self.dist[self.n]

                    if (next_v == -1 and self.dist[self.n] == INF) or \
                        (next_v != -1 and self.dist[next_v] ==
↪   INF):

                        target = next_v if next_v != -1 else self.n --
                        self.dist[target] = self.dist[v] + 1 ---------
                        if target != self.n: ------------------------
                            queue.append(target) --------------------

        return self.dist[self.n] != INF

    def dfs(self, v):
        if v != -1:
            for u in self.adj[v]:
                next_v = self.R[u]
                dist_next = self.dist[next_v] if next_v != -1 else
↪   self.dist[self.n]

                if dist_next == self.dist[v] + 1:
                    if self.dfs(next_v):
                        self.R[u] = v
                        self.L[v] = u
                        return True
            self.dist[v] = INF
            return False
        return True

    def maximum_matching(self):
        matching = 0
        for u in range(self.n):
            for v in self.adj[u]:
                if self.R[v] == -1:
                    self.L[u] = v
                    self.R[v] = u
                    matching += 1
                    break

        while self.bfs():
            for i in range(self.n):
                if self.L[i] == -1 and self.dfs(i):
                    matching += 1
        return matching
# Input Format: g = BipartiteGraph(2, 2)
#               g.add_edge(0, 0)
#               g.add_edge(0, 1)
#               g.add_edge(1, 0)
#               print(f"Max Matching: {g.maximum_matching()}")
```

4.2. Shortest Path.

4.2.1. *Dijkstra.*

```
import heapq
INF = float('inf')
def dijkstra(n, adj, s):
    dist = [INF] * n
    dist[s] = 0
    pq = [(0, s)]

    while pq:
        d, u = heapq.heappop(pq)
        if dist[u] < d:
            continue
        for v, w in adj[u]:
            if dist[v] > d + w:
                dist[v] = d + w
                heapq.heappush(pq, (dist[v], v))
    return dist
```

4.2.2. *Bellman Ford.*

```
INF = float('inf')
def bellman_ford(n, adj, s):
    dist = [INF] * n
    dist[s] = 0
    for _ in range(n - 1):
        for u in range(n):
            if dist[u] == INF: continue
            for v, w in adj[u]:
                if dist[u] + w < dist[v]:
                    dist[v] = dist[u] + w
    for u in range(n):
        if dist[u] == INF: continue
        for v, w in adj[u]:
            if dist[u] + w < dist[v]:
                return dist, True
    return dist, False
```

4.3. Minimum Spanning Tree.

4.3.1. *Kruskal Min Spanning Tree.*

```
import heapq

class UnionFind:
    def __init__(self, n):
        self.parent = list(range(n))
        self.rank = [0] * n

    def find(self, x):
        while self.parent[x] != x:
            self.parent[x] = self.parent[self.parent[x]]
            x = self.parent[x]
        return x

    def unite(self, a, b):
        a = self.find(a)
        b = self.find(b)

        if a == b:
            return False

        if self.rank[a] < self.rank[b]:
            self.parent[a] = b
        else:
            self.parent[b] = a
            if self.rank[a] == self.rank[b]:
                self.rank[a] += 1

        return True


def kruskal(n, edges):
    edges = edges[:]
    heapq.heapify(edges)

    uf = UnionFind(n)
    total_weight = 0
    mst = []

    while edges:
        w, u, v = heapq.heappop(edges)
        if uf.unite(u, v):
            mst.append((u, v, w))
            total_weight += w

            if len(mst) == n - 1:
                break

    return mst, total_weight
```

4.3.2. *Prim Min Spanning Tree.*

```
import heapq

def prim(n, adj, start=0):
    visited = [False] * n
    pq = [(0, start, -1)]

    total_weight = 0
    mst = []

    while pq:
        w, u, parent = heapq.heappop(pq)

        if visited[u]:
            continue

        visited[u] = True

        if parent != -1:
            mst.append((parent, u, w))
            total_weight += w

        for v, weight in adj[u]:
            if not visited[v]:
                heapq.heappush(pq, (weight, v, u))

    return mst, total_weight
```

4.4. Least Common Ancestor.

4.4.1. *Binary Lifting Lca.*

```
import math

class BinaryLifting:
    def __init__(self, adj):
        self.n = len(adj)
        self.LOG = math.ceil(math.log2(self.n)) + 1
        self.adj = adj

        self.dep = [0] * self.n
        self.par = [[-1] * self.LOG for _ in range(self.n)]

        self.build(0)

    def dfs(self, u, p):
        self.par[u][0] = p
        for v in self.adj[u]:
            if v != p:
                self.dep[v] = self.dep[u] + 1
                self.dfs(v, u)

    def build(self, root):
        self.dfs(root, root)

        for k in range(1, self.LOG):
            for u in range(self.n):
                self.par[u][k] = self.par[self.par[u][k-1]][k-1]

    def lift(self, u, k):
        for i in range(self.LOG):
            if k & (1 << i):
                u = self.par[u][i]
        return u

    def lca(self, u, v):
        if self.dep[u] < self.dep[v]:
            u, v = v, u

        u = self.lift(u, self.dep[u] - self.dep[v])

        if u == v:
            return u

        for k in reversed(range(self.LOG)):
            if self.par[u][k] != self.par[v][k]:
                u = self.par[u][k]
                v = self.par[v][k]

        return self.par[u][0]
```

4.4.2. *Euler Lca.*

```
class EulerLCA:
    def __init__(self, adj):
        self.n = len(adj)
        self.adj = adj

        self.euler = []
        self.first = [-1] * self.n
        self.depth = [0] * self.n

        self.build(0)

    def dfs(self, u, p, d):
        self.first[u] = len(self.euler)
        self.depth[u] = d
        self.euler.append(u)

        for v in self.adj[u]:
            if v != p:
                self.dfs(v, u, d + 1)
                self.euler.append(u)

    def build(self, root):
        self.dfs(root, root, 0)

        m = len(self.euler)
        LOG = m.bit_length()

        self.st = [[0] * m for _ in range(LOG)]
        self.log = [0] * (m + 1)

        for i in range(2, m + 1):
            self.log[i] = self.log[i // 2] + 1

        for i in range(m):
            self.st[0][i] = self.euler[i]

        for k in range(1, LOG):
            for i in range(m - (1 << k) + 1):
                a = self.st[k-1][i]
                b = self.st[k-1][i + (1 << (k-1))]
                self.st[k][i] = a if self.depth[a] < self.depth[b] else b

    def lca(self, u, v):
        l, r = self.first[u], self.first[v]
        if l > r:
            l, r = r, l

        k = self.log[r - l + 1]
        a = self.st[k][l]
        b = self.st[k][r - (1 << k) + 1]

        return a if self.depth[a] < self.depth[b] else b
```

4.4.3. *Tarjan Lca.*

```
class DSU:
    def __init__(self, n):
        self.p = list(range(n))

    def find(self, x):
        if self.p[x] != x:
```

```
        self.p[x] = self.find(self.p[x])
        return self.p[x]

    def unite(self, a, b):
        self.p[self.find(b)] = self.find(a)

class Tarjan:
    def __init__(self, adj, queries):
        self.adj = adj
        self.queries = queries
        self.n = len(adj)

        self.dsu = DSU(self.n)
        self.vis = [False] * self.n
        self.anc = [0] * self.n
        self.res = {}

    def dfs(self, u, p):
        self.anc[u] = u

        for v in self.adj[u]:
            if v == p:
                continue
            self.dfs(v, u)
            self.dsu.unite(u, v)
            self.anc[self.dsu.find(u)] = u

        self.vis[u] = True

        for v, idx in self.queries[u]:
            if self.vis[v]:
                self.res[idx] = self.anc[self.dsu.find(v)]
```

4.5. Minimum Mean Weight Cycle.

```
def min_mean_cycle(adj):
    n = len(adj)
    INF = float('inf')

    arr = [[INF] * n for _ in range(n + 1)]

    for i in range(n):
        arr[0][i] = 0

    for k in range(1, n + 1):
        for u in range(n):
            if arr[k - 1][u] == INF:
                continue
            for v, w in adj[u]:
                arr[k][v] = min(arr[k][v], arr[k - 1][u] + w)

    result = INF

    for v in range(n):
        if arr[n][v] == INF:
            continue

        max_avg = -INF
        for k in range(n):
            if arr[k][v] == INF:
                continue
            avg = (arr[n][v] - arr[k][v]) / (n - k)
            max_avg = max(max_avg, avg)

        result = min(result, max_avg)

    return result
```

4.6. Strongly Connected Components.

```
class KosarajuGraph:
    def __init__(self, n):
        self.n = n
        self.graph = [[] for _ in range(n)]
        self.rev_graph = [[] for _ in range(n)]
        self.vis = [0] * n
        self.sccs = []

    def add_edge(self, u, v):
        self.graph[u].append(v)
        self.rev_graph[v].append(u)

    def dfs(self, u, adj, container):
        self.vis[u] = 1
        for v in adj[u]:
            if not self.vis[v]:
                self.dfs(v, adj, container)
        container.append(u)

    def kosaraju(self):
        topo = []

        # First pass
        self.vis = [0] * self.n
        for i in range(self.n):
            if not self.vis[i]:
                self.dfs(i, self.graph, topo)

        # Second pass
        self.vis = [0] * self.n
        self.sccs = []

        for u in reversed(topo):
            if not self.vis[u]:
                comp = []
                self.dfs(u, self.rev_graph, comp)
                self.sccs.append(comp)

        return self.sccs
```

4.7. Heavy Light Decomposition.

```
class SegmentTree:
    def __init__(self, n):
        self.n = n
        self.tree = [0] * (4 * n)

    # Example: point update
    def update(self, idx, val, node=1, l=0, r=None):
        if r is None:
            r = self.n - 1
        if l == r:
            self.tree[node] = vala
            return
        mid = (l + r) // 2
        if idx <= mid:
            self.update(idx, val, node * 2, l, mid)
        else:
            self.update(idx, val, node * 2 + 1, mid + 1, r)
        self.tree[node] = self.tree[node * 2] + self.tree[node * 2 + 1]

    # Example: range query
    def query(self, ql, qr, node=1, l=0, r=None):
        if r is None:
            r = self.n - 1
        if qr < l or r < ql:
            return 0
        if ql <= l and r <= qr:
            return self.tree[node]
        mid = (l + r) // 2
        return self.query(ql, qr, node * 2, l, mid) + \
               self.query(ql, qr, node * 2 + 1, mid + 1, r)


class HeavyLightTree:
    def __init__(self, n):
        self.n = n
        self.adj = [[] for _ in range(n)]

        self.par = [-1] * n
        self.heavy = [-1] * n
        self.dep = [0] * n
        self.path_root = [0] * n
        self.pos = [0] * n

        self.segment_tree = SegmentTree(n)
        self.current_pos = 0

    def add_edge(self, u, v):
        self.adj[u].append(v)
        self.adj[v].append(u)

    def dfs(self, u):
        size = 1
        max_subtree_size = 0

        for v in self.adj[u]:
            if v != self.par[u]:
                self.par[v] = u
                self.dep[v] = self.dep[u] + 1
                subtree_size = self.dfs(v)

                if subtree_size > max_subtree_size:
                    max_subtree_size = subtree_size
                    self.heavy[u] = v

                size += subtree_size

        return size

    def decompose(self, u, root):
        self.path_root[u] = root
        self.pos[u] = self.current_pos
        self.current_pos += 1

        if self.heavy[u] != -1:
            self.decompose(self.heavy[u], root)

        for v in self.adj[u]:
            if v != self.par[u] and v != self.heavy[u]:
                self.decompose(v, v)

    def build(self, root=0):
        self.par[root] = root
        self.dep[root] = 0
        self.dfs(root)
        self.decompose(root, root)

    # Query path u -> v
    def query(self, u, v):
        res = 0
        while self.path_root[u] != self.path_root[v]:
            if self.dep[self.path_root[u]] <
↪   self.dep[self.path_root[v]]:
                u, v = v, u
            start = self.path_root[u]
            res += self.segment_tree.query(self.pos[start],
↪   self.pos[u])
            u = self.par[start]

        # Same path
        if self.dep[u] > self.dep[v]:
            u, v = v, u
        res += self.segment_tree.query(self.pos[u], self.pos[v])
        return res
```

```
    # Update node value
    def update(self, u, value):
        self.segment_tree.update(self.pos[u], value)
```

4.8. Euler Path/Cycle.

4.8.1. *Directed Euler.*

```
from collections import defaultdict

class DirectedEuler:
    def __init__(self, n):
        self.n = n
        self.adj = defaultdict(list)
        self.indeg = [0]*n
        self.outdeg = [0]*n
        self.edge_count = 0

    def add_edge(self, u, v):
        self.adj[u].append(v)
        self.outdeg[u] += 1
        self.indeg[v] += 1
        self.edge_count += 1

    def find_start_node(self):
        start = -1
        end = -1
        for i in range(self.n):
            if self.outdeg[i] - self.indeg[i] == 1:
                if start != -1:
                    return -1
                start = i
            elif self.indeg[i] - self.outdeg[i] == 1:
                if end != -1:
                    return -1
                end = i
            elif self.indeg[i] != self.outdeg[i]:
                return -1

        if start == -1:
            for i in range(self.n):
                if self.outdeg[i] > 0:
                    return i
        return start

    def get_euler_path(self):
        start = self.find_start_node()
        if start == -1:
            return None

        stack = [start]
        path = []

        local_adj = {u: self.adj[u][:] for u in self.adj}

        while stack:
            u = stack[-1]
            if u in local_adj and local_adj[u]:
                v = local_adj[u].pop()
                stack.append(v)
            else:
                path.append(stack.pop())

        if len(path) != self.edge_count + 1:
            return None

        return path[::-1]


    class UndirectedEuler:
        def __init__(self, n):
            self.n = n
            self.adj = defaultdict(list)
            self.edge_count = 0

        def add_edge(self, u, v):
            self.adj[u].append(v)
            self.adj[v].append(u)
            self.edge_count += 1

        def is_valid(self):
            odd = 0
            for i in range(self.n):
                if len(self.adj[i]) % 2 != 0:
                    odd += 1
            return odd == 0 or odd == 2

        def get_euler_path(self):
            if not self.is_valid():
                return None

            start = 0
            for i in range(self.n):
                if len(self.adj[i]) % 2 == 1:
                    start = i
                    break
                if len(self.adj[i]) > 0:
                    start = i

            stack = [start]
            path = []

            local_adj = {u: self.adj[u][:] for u in self.adj}

            while stack:
                u = stack[-1]
                if u in local_adj and local_adj[u]:
                    v = local_adj[u].pop()
                    local_adj[v].remove(u)
                    stack.append(v)
                else:
                    path.append(stack.pop())

            if len(path) != self.edge_count + 1:
                return None

            return path[::-1]
```

4.8.2. *Undirected Euler.*

```
from collections import defaultdict

class UndirectedEuler:
    def __init__(self, n):
        self.n = n
        self.adj = defaultdict(list)
        self.edge_count = 0

    def add_edge(self, u, v):
        self.adj[u].append(v)
        self.adj[v].append(u)
        self.edge_count += 1

    def is_valid(self):
        odd = 0
        for i in range(self.n):
            if len(self.adj[i]) % 2 != 0:
                odd += 1
        return odd == 0 or odd == 2

    def get_euler_path(self):
        if not self.is_valid():
            return None

        start = 0
        for i in range(self.n):
            if len(self.adj[i]) % 2 == 1:
                start = i
                break
            if len(self.adj[i]) > 0:
                start = i

        stack = [start]
        path = []

        local_adj = {u: self.adj[u][:] for u in self.adj}

        while stack:
            u = stack[-1]
            if u in local_adj and local_adj[u]:
                v = local_adj[u].pop()
                local_adj[v].remove(u)
                stack.append(v)
            else:
                path.append(stack.pop())

        if len(path) != self.edge_count + 1:
            return None

        return path[::-1]
```

4.9. Tree Isomorphism.

```
from collections import deque

def bfs(u, adj, n):
    vis = [False] * n
    pre = [-1] * n
    q = deque()

    q.append(u)
    vis[u] = True

    last = u

    while q:
        u = q.popleft()
        last = u

        for v in adj[u]:
            if not vis[v]:
                vis[v] = True
                pre[v] = u
                q.append(v)

    return last, pre

def tree_centers(r, adj, n):
    u, _ = bfs(r, adj, n)
    v, pre = bfs(u, adj, n)

    path = []
    while v != -1:
        path.append(v)
        v = pre[v]

    size = len(path)

    med = [path[size // 2]]
    if size % 2 == 0:
        med.append(path[size // 2 - 1])

    return med

def rootcode(u, adj, p, d, pr):
```

```
    children = []
    for v in adj[u]:
        if v != p:
            children.append(rootcode(v, adj, u, (d + 1) % len(pr), pr))

    children.sort()

    h = len(children) + 1
    for x in children:
        h = h * pr[d] + x

    return h

def treecode(root, adj, n, pr):
    c = tree_centers(root, adj, n)

    if len(c) == 1:
        return (rootcode(c[0], adj, -1, 0, pr) << 1) | 1
    else:
        return (rootcode(c[0], adj, -1, 0, pr) *
                rootcode(c[1], adj, -1, 0, pr)) << 1

def isomorphic(r1, adj1, r2, adj2, n1, n2, pr, rooted=False):
    if rooted:
        return rootcode(r1, adj1, -1, 0, pr) == rootcode(r2, adj2, -1,
↪   0, pr)
    return treecode(r1, adj1, n1, pr) == treecode(r2, adj2, n2, pr) --
```

4.10. Centroid Decomposition.

```
import sys
sys.setrecursionlimit(10**7)

INF = 10**9

class CentroidDecomposition:
    def __init__(self, n):
        self.n = n
        self.adj = [[] for _ in range(n)]

        self.LG = 20
        self.jmp = [[-1] * self.LG for _ in range(n)]
        self.path = [[INF] * self.LG for _ in range(n)]

        self.sz = [0] * n
        self.seph = [0] * n
        self.shortest = [INF] * n
        self.blocked = [False] * n

    def add_edge(self, a, b):
        self.adj[a].append(b)
        self.adj[b].append(a)

    def dfs(self, u, p):
        self.sz[u] = 1
        for v in self.adj[u]:
            if v != p and not self.blocked[v]:
                self.sz[u] += self.dfs(v, u)
        return self.sz[u]

    def find_centroid(self, u, p, n):
        for v in self.adj[u]:
            if v != p and not self.blocked[v] and self.sz[v] > n // 2:
                return self.find_centroid(v, u, n)
        return u

    def makepaths(self, sep, u, p, depth, h):
        self.jmp[u][h] = sep
        self.path[u][h] = depth

        for v in self.adj[u]:
            if v != p and not self.blocked[v]:
                self.makepaths(sep, v, u, depth + 1, h)

    def separate(self, h=0, u=0):
        self.dfs(u, -1)
        c = self.find_centroid(u, -1, self.sz[u])

        self.seph[c] = h
        self.makepaths(c, c, -1, 0, h)

        self.blocked[c] = True

        for v in self.adj[c]:
            if not self.blocked[v]:
                self.separate(h + 1, v)

    def paint(self, u):
        for h in range(self.seph[u] + 1):
            c = self.jmp[u][h]
            self.shortest[c] = min(self.shortest[c], self.path[u][h])

    def closest(self, u):
        ans = INF
        for h in range(self.seph[u] + 1):
            c = self.jmp[u][h]
            ans = min(ans, self.shortest[c] + self.path[u][h])
        return ans
```

5. Math I Algebra

5.0.1. *Generating Function Manager.*

```
MOD = 998244353

def mod_pow(base, exp):
    res = 1
    base %= MOD
    while exp:
        if exp & 1:
            res = res * base % MOD
        base = base * base % MOD
        exp >>= 1
    return res

class GF_Manager:
    DEPTH = 23

    def __init__(self):
        self.prim = [0] * (self.DEPTH + 1)
        self.prim_inv = [0] * (self.DEPTH + 1)
        self.two_inv = [0] * (self.DEPTH + 1)
        self.set_up_primitives()

    def set_up_primitives(self):
        self.prim[self.DEPTH] = 31
        self.prim_inv[self.DEPTH] = mod_pow(self.prim[self.DEPTH], MOD
↪   - 2)
        self.two_inv[self.DEPTH] = mod_pow(1 << self.DEPTH, MOD - 2) -

        for n in range(self.DEPTH - 1, -1, -1):
            self.prim[n] = self.prim[n + 1] * self.prim[n + 1] % MOD -
            self.prim_inv[n] = mod_pow(self.prim[n], MOD - 2)
            self.two_inv[n] = mod_pow(1 << n, MOD - 2)

    def NTT(self, A, n, is_inverse=False):
        if n == 0:
            return A[:]

        half = 1 << (n - 1)
        even = A[0::2]
        odd = A[1::2]

        even = self.NTT(even, n - 1, is_inverse)
        odd = self.NTT(odd, n - 1, is_inverse)

        res = [0] * (1 << n)
        w1 = self.prim_inv[n] if is_inverse else self.prim[n]
        w = 1

        for i in range(half):
            t = w * odd[i] % MOD
            res[i] = (even[i] + t) % MOD
            res[i + half] = (even[i] - t) % MOD
            w = w * w1 % MOD

        return res

    def mult(self, A, B):
        an, bn = len(A), len(B)
        degree = an + bn - 1

        n = 0
        while (1 << n) < degree:
            n += 1

        size = 1 << n
        tA = A + [0] * (size - an)
        tB = B + [0] * (size - bn)

        tA = self.NTT(tA, n)
        tB = self.NTT(tB, n)

        for i in range(size):
            tA[i] = tA[i] * tB[i] % MOD

        tA = self.NTT(tA, n, True)

        inv = self.two_inv[n]
        return [(x * inv) % MOD for x in tA[:degree]]

    def horners(self, F, x):
        ans = 0
        for coef in reversed(F):
            ans = (ans * x + coef) % MOD
        return ans
```

5.0.2. *Fast Fourier Transform.*

```
import math

class Poly:
    def __init__(self, a=0.0, b=0.0):
        self.a = a
        self.b = b

    def __add__(self, other):
        return Poly(self.a + other.a, self.b + other.b)

    def __sub__(self, other):
        return Poly(self.a - other.a, self.b - other.b)

    def __mul__(self, other):
        return Poly(
            self.a * other.a - self.b * other.b,
            self.a * other.b + self.b * other.a
        )

    def __repr__(self):
        return f"({self.a}+{self.b}i)"

def fft_recursive(inp, n, s):
    if n == 1:
        return [inp[0]]
```

```
--- half = n // 2
--- even = fft_recursive(inp, half, s * 2)
--- odd = fft_recursive(inp[s:], half, s * 2)

--- res = [None] * n
--- w = Poly(1, 0)
--- wn = Poly(math.cos(math.pi / half), math.sin(math.pi / half))

--- for i in range(half):
------- t = w * odd[i]
------- res[i] = even[i] + t
------- res[i + half] = even[i] - t
------- w = w * wn

--- return res

def fft(p):
--- return fft_recursive(p, len(p), 1)

def inverse_fft(p):
--- n = len(p)

--- p = [Poly(x.a, -x.b) for x in p]

--- p = fft(p)

--- res = []
--- for x in p:
------- res.append(Poly(x.a / n, -x.b / n))

--- return res
```

5.0.3. *Fft Polynomial Multiplication.*

```
def fft_multiply(a, b):
--- an, bn = len(a), len(b)
--- degree = an + bn - 1

--- n = 1
--- while n < degree:
------- n <<= 1

--- A = [Poly(x, 0) for x in a] + [Poly(0, 0)] * (n - an)
--- B = [Poly(x, 0) for x in b] + [Poly(0, 0)] * (n - bn)

--- A = fft(A)
--- B = fft(B)

--- for i in range(n):
------- A[i] = A[i] * B[i]

--- A = inverse_fft(A)

--- result = [int(round(A[i].a)) for i in range(degree)]
--- return result
```

5.0.4. *Number Theoretic Transform.*

```
MOD = 998244353
PRIMITIVE_ROOT = 3

def mod_pow(base, exp, mod=MOD):
--- res = 1
--- base %= mod
--- while exp:
------- if exp & 1:
----------- res = res * base % mod
------- base = base * base % mod
------- exp >>= 1
--- return res

class Num:
--- def __init__(self, x=0):
------- self.x = x % MOD

--- def __add__(self, other):
------- return Num(self.x + other.x)

--- def __sub__(self, other):
------- return Num(self.x - other.x)

--- def __mul__(self, other):
------- return Num(self.x * other.x)

--- def __truediv__(self, other):
------- return self * other.inv()

--- def inv(self):
------- return Num(mod_pow(self.x, MOD - 2))

--- def pow(self, p):
------- return Num(mod_pow(self.x, p))

--- def __repr__(self):
------- return str(self.x)

def ntt(a, invert=False):
--- n = len(a)

--- j = 0
--- for i in range(1, n):
------- bit = n >> 1
------- while j & bit:
----------- j ^= bit
----------- bit >>= 1
------- j |= bit
------- if i < j:
----------- a[i], a[j] = a[j], a[i]

--- length = 2
--- while length <= n:
------- wlen = mod_pow(PRIMITIVE_ROOT, (MOD - 1) // length)
------- if invert:
----------- wlen = mod_pow(wlen, MOD - 2)

------- for i in range(0, n, length):
----------- w = 1
----------- half = length // 2
----------- for j in range(i, i + half):
--------------- u = a[j]
--------------- v = Num(a[j + half].x * w % MOD)
--------------- a[j] = u + v
--------------- a[j + half] = u - v
--------------- w = w * wlen % MOD

------- length <<= 1

--- if invert:
------- inv_n = mod_pow(n, MOD - 2)
------- for i in range(n):
----------- a[i] = Num(a[i].x * inv_n % MOD)

def multiply(A, B):
--- n = 1
--- while n < len(A) + len(B):
------- n <<= 1

--- fa = [Num(x) for x in A] + [Num(0)] * (n - len(A))
--- fb = [Num(x) for x in B] + [Num(0)] * (n - len(B))

--- ntt(fa)
--- ntt(fb)

--- for i in range(n):
------- fa[i] = fa[i] * fb[i]

--- ntt(fa, True)

--- return [fa[i].x for i in range(len(A) + len(B) - 1)]

def poly_inv(x, l):
--- if l == 1:
------- return [mod_pow(x[0], MOD - 2)]

--- y = poly_inv(x, (l + 1) // 2)

--- n = 1
--- while n < 2 * l:
------- n <<= 1

--- A = x[:l] + [0] * (n - l)
--- B = y[:] + [0] * (n - len(y))

--- A = [Num(v) for v in A]
--- B = [Num(v) for v in B]

--- ntt(A)
--- ntt(B)

--- for i in range(n):
------- B[i] = B[i] * (Num(2) - A[i] * B[i])

--- ntt(B, True)

--- return [B[i].x % MOD for i in range(l)]

def poly_sqrt(x, l):
--- if l == 1:
------- assert x[0] == 1
------- return [1]

--- y = poly_sqrt(x, l // 2)
--- inv_y = poly_inv(y, l // 2)

--- n = 1
--- while n < 2 * l:
------- n <<= 1

--- T1 = x[:l] + [0] * (n - l)
--- T2 = inv_y[:] + [0] * (n - len(inv_y))

--- T1 = [Num(v) for v in T1]
--- T2 = [Num(v) for v in T2]

--- ntt(T1)
--- ntt(T2)

--- for i in range(n):
------- T2[i] = T1[i] * T2[i]

--- ntt(T2, True)

--- inv2 = (MOD + 1) // 2
--- res = []
--- for i in range(l):
------- val = (y[i] + T2[i].x) * inv2 % MOD
------- res.append(val)

--- return res
```

5.0.5. *Polynomial Long Division.*

```
MOD = 998244353
G = 3

def mod_pow(a, e):
    r = 1
    while e:
        if e & 1:
            r = r * a % MOD
        a = a * a % MOD
        e >>= 1
    return r

def bit_reverse(a):
    n = len(a)
    j = 0
    for i in range(1, n):
        bit = n >> 1
        while j & bit:
            j ^= bit
            bit >>= 1
        j ^= bit
        if i < j:
            a[i], a[j] = a[j], a[i]

def ntt(a, invert=False):
    n = len(a)
    bit_reverse(a)

    length = 2
    while length <= n:
        wlen = mod_pow(G, (MOD - 1) // length)
        if invert:
            wlen = mod_pow(wlen, MOD - 2)

        for i in range(0, n, length):
            w = 1
            half = length // 2
            for j in range(i, i + half):
                u = a[j]
                v = a[j + half] * w % MOD
                a[j] = (u + v) % MOD
                a[j + half] = (u - v) % MOD
                w = w * wlen % MOD

        length <<= 1

    if invert:
        inv_n = mod_pow(n, MOD - 2)
        for i in range(n):
            a[i] = a[i] * inv_n % MOD

def multiply(a, b):
    n = 1
    while n < len(a) + len(b):
        n <<= 1

    fa = a[:] + [0] * (n - len(a))
    fb = b[:] + [0] * (n - len(b))

    ntt(fa)
    ntt(fb)

    for i in range(n):
        fa[i] = fa[i] * fb[i] % MOD

    ntt(fa, True)

    return fa[:len(a) + len(b) - 1]

def poly_inv(f, n):
    if n == 1:
        return [mod_pow(f[0], MOD - 2)]

    m = (n + 1) // 2
    g = poly_inv(f, m)

    size = 1
    while size < 2 * n:
        size <<= 1

    a = f[:n] + [0] * (size - len(f[:n]))
    g = g + [0] * (size - len(g))

    ntt(a)
    ntt(g)

    for i in range(size):
        g[i] = g[i] * (2 - a[i] * g[i] % MOD) % MOD

    ntt(g, True)

    return g[:n]

def poly_divide(A, B):
    n, m = len(A), len(B)

    if n < m:
        return [], A[:]

    k = n - m + 1

    A_rev = A[::-1]
    B_rev = B[::-1]

    B_inv = poly_inv(B_rev, k)

    Q_rev = multiply(A_rev, B_inv)[:k]
    Q = Q_rev[::-1]

    BQ = multiply(B, Q)

    R = [(A[i] - (BQ[i] if i < len(BQ) else 0)) % MOD for i in
    ↪  range(n)]

    while Q and Q[-1] == 0:
        Q.pop()
    while R and R[-1] == 0:
        R.pop()

    return Q, R
```

5.0.6. *Matrix Multiplication.*

```
MOD = 998244353

def multiply(A, B):
    p = len(A)
    q = len(A[0])
    r = len(B[0])

    if len(B) != q:
        raise ValueError("Dimension mismatch")

    AB = [[0] * r for _ in range(p)]

    for i in range(p):
        for j in range(q):
            for k in range(r):
                AB[i][k] = (AB[i][k] + A[i][j] * B[j][k]) % MOD

    return AB
```

5.0.7. *Matrix Power.*

```
MOD = 998244353

def multiply(A, B):
    p, q, r = len(A), len(A[0]), len(B[0])

    res = [[0] * r for _ in range(p)]

    for i in range(p):
        for j in range(q):
            for k in range(r):
                res[i][k] = (res[i][k] + A[i][j] * B[j][k]) % MOD

    return res

def identity(n):
    I = [[0] * n for _ in range(n)]
    for i in range(n):
        I[i][i] = 1
    return I

def power(B, e):
    n = len(B)

    result = identity(n)
    base = [row[:] for row in B]

    while e > 0:
        if e & 1:
            result = multiply(result, base)
        base = multiply(base, base)
        e >>= 1

    return result
```

5.0.8. *Fibonacci Matrix.*

```
EPS = 1e-9

def gauss_jordan(A):
    n = len(A)
    m = len(A[0])

    singular = False

    i = 0
    for p in range(m):
        if i >= n:
            break

        pivot = i
        for k in range(i + 1, n):
            if abs(A[k][p]) > abs(A[pivot][p]):
                pivot = k

        if abs(A[pivot][p]) < EPS:
            singular = True
            continue

        A[i], A[pivot] = A[pivot], A[i]

        div = A[i][p]
        for j in range(p, m):
            A[i][j] /= div

        for k in range(n):
            if k == i:
                continue
            factor = A[k][p]
            for j in range(p, m):
                A[k][j] -= factor * A[i][j]

        i += 1
```

```
--- return not singular

def apply(A, x):
--- return [sum(A[i][j] * x[j] for j in range(len(x))) for i in range(len(A))]
```

6. Math II Combinatorics

6.0.1. *Lucas Theorem.*

```
_fact_cache = []
_last_p = -1

def lucas(n, k, p):
--- if k < 0 or k > n:
------- return 0
--- if k == 0 or n == k:
------- return 1

--- global _fact_cache, _last_p
--- if _last_p != p:
------- _fact_cache = [1] * p
------- for i in range(2, p):
----------- _fact_cache[i] = (_fact_cache[i - 1] * i) % p
------- _last_p = p

--- res_num = 1
--- res_den = 1

--- while n > 0 or k > 0:
------- ni, ki = n % p, k % p
------- if ki > ni:
----------- return 0
------- res_num = (res_num * _fact_cache[ni]) % p
------- res_den = (res_den * _fact_cache[ki] * _fact_cache[ni - ki]) % p
------- n //= p
------- k //= p
--- return (res_num * pow(res_den, p - 2, p)) % p
# Input Format: print(lucas(n, k, p))
```

6.0.2. *Catalan Numbers.*

```
def catalan(n):
--- if (n == 0 or n == 1):
------- return 1

--- dp = [0] * (n + 1)
--- dp[0] = dp[1] = 1

--- for i in range(2, n + 1):
------- for j in range(i):
----------- dp[i] += dp[j] * dp[i-j-1]

--- return dp[n]

# for i in range(10):
#     print(catalan(i))
```

6.0.3. *Kth Permutation.*

```
#fact = [1, 1, 2, 6, 24, 120, 720, 5040, 40320, 362880]

def findKthPermutation(n, k):
--- nums = [str(i) for i in range(1, n + 1)]

--- fact = [1] * n
--- for i in range(1, n):
------- fact[i] = fact[i - 1] * i

--- k -= 1
--- res = []
--- for i in range(n - 1, -1, -1):
------- idx = k // fact[i]
------- k %= fact[i]

------- res.append(nums.pop(idx))

--- return ''.join(res)
```

6.0.4. *Factoradics.*

```
def factoradic(arr):
--- n = len(arr)
--- ft = FenwickTree(n)
--- for i in range(1, n + 1):
------- ft.add(i, 1)

--- res = []
--- for val in arr:
------- bit_val = val + 1
------- s = ft.query(bit_val - 1)
------- ft.add(bit_val, -1)
------- res.append(s)
--- return res

def permute(fact, n):
--- ft = FenwickTree(n)
--- for i in range(1, n + 1):
------- ft.add(i, 1)

--- res = []
--- for val in fact:
------- idx = ft.find_kth(val)
------- ft.add(idx, -1)
------- res.append(idx - 1)
--- return res
```

7. Math III Number Theory

7.0.1. *Gcd Subset Counter.*

```
def gcd_subset_count(arr):
--- n = len(arr)
--- mx = max(arr)

--- f = [0] * (mx + 1)
--- for x in arr:
------- f[x] += 1

--- gcnt = [0] * (mx + 1)

--- for i in range(mx, 0, -1):
------- add = 0

------- for j in range(i, mx + 1, i):
----------- add += f[j]

------- total = (1 << add) - 1  # 2^add - 1

------- sub = 0
------- for j in range(2 * i, mx + 1, i):
----------- sub += gcnt[j]

------- gcnt[i] = total - sub

--- return gcnt
```

7.0.2. *Linear Diophantine.*

```
def gcd(a, b):
--- if a == 0:
------- return b
--- return gcd(b % a, a)

def isPossible(a, b, c):
--- g = gcd(a, b)

--- return c % g == 0
```

7.0.3. *Chinese Remainder Theorem.*

```
def gcd_extended(a, b):
--- if a == 0:
------- return b, 0, 1
--- gcd, x1, y1 = gcd_extended(b % a, a)
--- x = y1 - (b // a) * x1
--- y = x1
--- return gcd, x, y

def find_min_x(num, rem):
--- prod = 1
--- for n in num:
------- prod *= n

--- result = 0
--- for i in range(len(num)):
------- prod_i = prod // num[i]
------- _, inv_i, _ = gcd_extended(prod_i, num[i])
------- result += rem[i] * prod_i * inv_i

--- return result % prod
```

7.0.4. *Euler Totient.*

```
def totient(n):
--- if n <= 1:
------- return 1

--- tot = n
--- i = 2

--- while i * i <= n:
------- if n % i == 0:
----------- tot -= tot // i
----------- while n % i == 0:
--------------- n //= i
------- i += 1

--- if n > 1:
------- tot -= tot // n

--- return tot
```

7.0.5. *Extended Euclidean Algorithm.*

```
def mod(x, m):
--- if m == 0:
------- return 0
--- if m < 0:
------- m = -m
--- return (x % m + m) % m

def extended_euclid(a, b):
--- if b == 0:
------- return a, 1, 0

--- g, x1, y1 = extended_euclid(b, a % b)

--- x = y1
--- y = x1 - (a // b) * y1

--- return g, x, y
```

7.0.6. *Modular Exponentiation.*

```
def mod_pow(b, e, m):
    res = 1 % m

    while e > 0:
        if e & 1:
            res = (res * b) % m
        b = (b * b) % m
        e >>= 1

    return res
```

7.0.7. *Mod Inverse.*

```
def extended_euclid(a, b):
    if b == 0:
        return a, 1, 0

    g, x1, y1 = extended_euclid(b, a % b)

    x = y1
    y = x1 - (a // b) * y1

    return g, x, y


def modinv(a, m):
    g, x, y = extended_euclid(a, m)

    if g != 1:
        return 0  # no inverse

    return x % m
```

7.0.8. *Modulo Solver.*

```
def extended_euclid(a, b):
    if b == 0:
        return a, 1, 0

    g, x1, y1 = extended_euclid(b, a % b)

    x = y1
    y = x1 - (a // b) * y1

    return g, x, y


def modsolver(a, b, m):
    g, x, y = extended_euclid(a, m)

    if b % g != 0:
        return (-1, -1)

    m_reduced = m // g

    x0 = (x * (b // g)) % m_reduced

    return (x0, m_reduced)
```

7.0.9. *Primitive Root.*

```
def primitive_root(m):
    if m == 2:
        return 1

    phi = m - 1

    factors = set()
    x = phi
    i = 2

    while i * i <= x:
        if x % i == 0:
            factors.add(i)
            while x % i == 0:
                x //= i
        i += 1

    if x > 1:
        factors.add(x)

    for g in range(2, m):
        ok = True
        for f in factors:
            if mod_pow(g, phi // f, m) == 1:
                ok = False
                break
        if ok:
            return g

    return -1
```

7.0.10. *Josephus.*

```
def josephus(n, k):
    res = 0
    for i in range(2, n + 1):
        res = (res + k) % i
    return res
```

8. Math IV Numerical Methods

8.0.1. *Fast Square Testing.*

```
import math

class FastSquareTester:
    def __init__(self):
        self.M = 0
        self.init_is_square()

    def init_is_square(self):
        for i in range(64):
            self.M |= (1 << (i * i % 64))

    def is_square(self, x):
        if x < 0: return False
        if x == 0: return True

        if not (self.M & (1 << (x & 63))):
            return False

        c = (x & -x).bit_length() - 1
        if c & 1:
            return False

        temp_x = x >> c
        if (temp_x & 7) != 1:
            return False

        r = int(math.isqrt(temp_x))
        return r * r == temp_x

tester = FastSquareTester()
```

8.0.2. *Simpson Integration.*

```
def f(x):
    return x ** 2
def simpson_integration(a, b, n=1000000):
    if n % 2 != 0:
        n += 1

    h = (b - a) / n
    s = f(a) + f(b)

    for i in range(1, n):
        x = a + h * i
        if i % 2 == 1:
            s += 4 * f(x)
        else:
            s += 2 * f(x)

    return s * (h / 3)
```

9. Strings

9.0.1. *Longest Common Prefix.*

```
def build_lcp(s, sa):
    n = len(s)
    rank = [0] * n
    lcp = [0] * n

    for i in range(n):
        rank[sa[i]] = i

    k = 0
    for i in range(n):
        if rank[i] == 0:
            k = 0
            continue

        j = sa[rank[i] - 1]

        while i + k < n and j + k < n and s[i + k] == s[j + k]:
            k += 1

        lcp[rank[i]] = k

        if k > 0:
            k -= 1

    return lcp
```

9.0.2. *Hasher.*

```
class Hasher:
    def __init__(self, s, bases=[31, 37], mod=10 ** 9 + 7):
        self.bases = bases
        self.mod = mod
        self.n = len(s)
        self.num_hashes = len(bases)
        self.p_pow = [[1] * (self.n + 1) for _ in range(self.num_hashes)]
        self.h_ans = [[0] * (self.n + 1) for _ in range(self.num_hashes)]

        for i, b in enumerate(self.bases):
            for j in range(self.n):
                self.p_pow[i][j + 1] = (self.p_pow[i][j] * b) % self.mod
                self.h_ans[i][j + 1] = (self.h_ans[i][j] + s[j] * self.p_pow[i][j]) % self.mod

    def get_hash(self, l, r):
        return tuple((self.h_ans[i][r + 1] - self.h_ans[i][l]) % self.mod
                     for i in range(self.num_hashes))

    def are_equal(self, l1, r1, l2, r2):
        if (r1 - l1) != (r2 - l2): return False

        h1 = self.get_hash(l1, r1)
        h2 = self.get_hash(l2, r2)

        return all((h1[i] * self.p_pow[i][l2]) % self.mod ==
```

```
------------------ (h2[i] * self.p_pow[i][l1]) % self.mod ------------
------------------ for i in range(self.num_hashes)) ------------------
```

9.0.3. *Z Algorithm.*

9.0.4. *Aho Corasick Trie.*

```
from collections import deque ----------------------------------------
import random --------------------------------------------------------
import string --------------------------------------------------------
 ---------------------------------------------------------------------
class Node: ----------------------------------------------------------
--- def __init__(self): ------------------------------------------
------- self.next = {} ---------------------------------------------
------- self.fail = None -------------------------------------------
------- self.count = 0 ---------------------------------------------
 ---------------------------------------------------------------------
--- def add(self, s): --------------------------------------------
------- node = self ------------------------------------------------
------- for c in s: ------------------------------------------------
----------- if c not in node.next: -------------------------------
--------------- node.next[c] = Node() ------------------------------
----------- node = node.next[c] ----------------------------------
------- node.count += 1 --------------------------------------------
 ---------------------------------------------------------------------
--- def prepare(self): -------------------------------------------
------- root = self ------------------------------------------------
------- q = deque() ------------------------------------------------
 ---------------------------------------------------------------------
------- root.fail = root -------------------------------------------
 ---------------------------------------------------------------------
------- for child in root.next.values(): ---------------------------
----------- child.fail = root ------------------------------------
----------- q.append(child) --------------------------------------
 ---------------------------------------------------------------------
------- while q: ---------------------------------------------------
----------- head = q.popleft() -----------------------------------
 ---------------------------------------------------------------------
----------- for c, nxt in head.next.items(): ---------------------
--------------- f = head.fail --------------------------------------
 ---------------------------------------------------------------------
--------------- while f != root and c not in f.next: -----------------
------------------- f = f.fail -----------------------------------
 ---------------------------------------------------------------------
--------------- nxt.fail = f.next[c] if c in f.next else root --------
 ---------------------------------------------------------------------
--------------- nxt.count += nxt.fail.count --------------------------
 ---------------------------------------------------------------------
--------------- q.append(nxt) ----------------------------------------
 ---------------------------------------------------------------------
--- def search(self, s): ----------------------------------------
------- root = self ------------------------------------------------
------- p = self ---------------------------------------------------
------- ans = 0 ----------------------------------------------------
 ---------------------------------------------------------------------
------- for c in s: ------------------------------------------------
----------- while p != root and c not in p.next: -----------------
--------------- p = p.fail -----------------------------------------
 ---------------------------------------------------------------------
----------- if c in p.next: --------------------------------------
--------------- p = p.next[c] --------------------------------------
--------------- ans += p.count -------------------------------------
 ---------------------------------------------------------------------
------- return ans -------------------------------------------------
```

9.0.5. *Knuth Morris Pratt.*

```
def build_kmp(f): ----------------------------------------------------
--- n = len(f) ---------------------------------------------------
--- par = [0] * (n + 1) ------------------------------------------
 ---------------------------------------------------------------------
--- par[0] = -1 --------------------------------------------------
--- if n > 0: ----------------------------------------------------
------- par[1] = 0 -------------------------------------------------
 ---------------------------------------------------------------------
--- i, j = 2, 0 --------------------------------------------------
 ---------------------------------------------------------------------
--- while i <= n: ------------------------------------------------
------- if f[i - 1] == f[j]: ---------------------------------------
----------- j += 1 -----------------------------------------------
----------- par[i] = j -------------------------------------------
----------- i += 1 -----------------------------------------------
------- elif j > 0: ------------------------------------------------
----------- j = par[j] -------------------------------------------
------- else: ------------------------------------------------------
----------- par[i] = 0 -------------------------------------------
----------- i += 1 -----------------------------------------------
 ---------------------------------------------------------------------
--- return par ---------------------------------------------------
 ---------------------------------------------------------------------
 ---------------------------------------------------------------------
def kmp(s, f): -------------------------------------------------------
--- par = build_kmp(f) -------------------------------------------
 ---------------------------------------------------------------------
--- i = 0 --------------------------------------------------------
--- j = 0 --------------------------------------------------------
--- ans = [] -----------------------------------------------------
 ---------------------------------------------------------------------
--- n, m = len(s), len(f) ----------------------------------------
 ---------------------------------------------------------------------
--- while i + j < n: ---------------------------------------------
------- if s[i + j] == f[j]: ---------------------------------------
----------- j += 1 -----------------------------------------------
----------- if j == m: -------------------------------------------
--------------- ans.append(i) --------------------------------------
--------------- i += j - par[j] ------------------------------------
--------------- if j > 0: ------------------------------------------
------------------- j = par[j] -----------------------------------
------- else: ------------------------------------------------------
----------- i += j - par[j] --------------------------------------
----------- if j > 0: --------------------------------------------
--------------- j = par[j] -----------------------------------------
 ---------------------------------------------------------------------
--- return ans ---------------------------------------------------
```

9.0.6. *Trie.*

```
class Node: ----------------------------------------------------------
--- def __init__(self): ------------------------------------------
------- self.children = {} -----------------------------------------
------- self.prefixes = 0 ------------------------------------------
------- self.words = 0 ---------------------------------------------
 ---------------------------------------------------------------------
class Trie: ----------------------------------------------------------
--- def __init__(self): ------------------------------------------
------- self.root = Node() -----------------------------------------
 ---------------------------------------------------------------------
--- def insert(self, seq): ---------------------------------------
------- cur = self.root --------------------------------------------
 ---------------------------------------------------------------------
------- for i in range(len(seq) + 1): ------------------------------
----------- cur.prefixes += 1 ------------------------------------
 ---------------------------------------------------------------------
----------- if i == len(seq): ------------------------------------
--------------- cur.words += 1 -------------------------------------
--------------- break ----------------------------------------------
 ---------------------------------------------------------------------
----------- head = seq[i] ----------------------------------------
 ---------------------------------------------------------------------
----------- if head not in cur.children: -------------------------
--------------- cur.children[head] = Node() ------------------------
 ---------------------------------------------------------------------
----------- cur = cur.children[head] -----------------------------
 ---------------------------------------------------------------------
--- def countMatches(self, seq): ---------------------------------
------- cur = self.root --------------------------------------------
 ---------------------------------------------------------------------
------- for c in seq: ----------------------------------------------
----------- if c not in cur.children: ----------------------------
--------------- return 0 ------------------------------------------
----------- cur = cur.children[c] --------------------------------
 ---------------------------------------------------------------------
------- return cur.words -------------------------------------------
 ---------------------------------------------------------------------
--- def countPrefixes(self, seq): ------------------------------------
------- cur = self.root --------------------------------------------
 ---------------------------------------------------------------------
------- for c in seq: ----------------------------------------------
----------- if c not in cur.children: ----------------------------
--------------- return 0 ------------------------------------------
----------- cur = cur.children[c] --------------------------------
 ---------------------------------------------------------------------
------- return cur.prefixes ----------------------------------------
```

9.0.7. *Suffix Array.*

```
class SuffixArray: ---------------------------------------------------
--- def __init__(self): ------------------------------------------
------- self.n = 0 -------------------------------------------------
------- self.suffix = [] -------------------------------------------
------- self.equiv = [] --------------------------------------------
------- self.equiv_pair = [] ---------------------------------------
------- self.s = "" ------------------------------------------------
 ---------------------------------------------------------------------
--- def build(self, s): ------------------------------------------
------- if not s.endswith('$'): ------------------------------------
----------- s += '$' ---------------------------------------------
 ---------------------------------------------------------------------
------- self.s = s -------------------------------------------------
------- self.n = len(s) --------------------------------------------
 ---------------------------------------------------------------------
------- self.suffix = list(range(self.n)) --------------------------
------- self.equiv = [0] * self.n ----------------------------------
------- self.equiv_pair = [(0, 0)] * self.n ------------------------
 ---------------------------------------------------------------------
------- self.suffix.sort(key=lambda i: s[i]) -----------------------
 ---------------------------------------------------------------------
------- cls = 0 ----------------------------------------------------
------- for i in range(self.n): ------------------------------------
----------- if i == 0 or s[self.suffix[i]] != s[self.suffix[i - 1]]: -
--------------- cls += 1 -------------------------------------------
----------- self.equiv[self.suffix[i]] = cls ---------------------
 ---------------------------------------------------------------------
------- k = 1 ------------------------------------------------------
------- while k < self.n: ------------------------------------------
----------- for i in range(self.n): ------------------------------
--------------- self.equiv_pair[i] = ( -----------------------------
------------------- self.equiv[i], -------------------------------
------------------- self.equiv[(i + k) % self.n] -----------------
--------------- ) --------------------------------------------------
 ---------------------------------------------------------------------
----------- self.suffix.sort(key=lambda i: self.equiv_pair[i]) -------
 ---------------------------------------------------------------------
----------- cls = 0 ----------------------------------------------
----------- new_equiv = [0] * self.n -----------------------------
 ---------------------------------------------------------------------
----------- for i in range(self.n): ------------------------------
--------------- if i == 0 or self.equiv_pair[self.suffix[i]] !=
↪  self.equiv_pair[self.suffix[i - 1]]:
------------------- cls += 1 -------------------------------------
--------------- new_equiv[self.suffix[i]] = cls ----------------------
 ---------------------------------------------------------------------
----------- self.equiv = new_equiv -------------------------------
----------- k <<= 1 ----------------------------------------------
 ---------------------------------------------------------------------
--- def _cmp(self, start, pattern): ------------------------------
```

```
------- s = self.s
------- n = len(s)

------- i = start
------- j = 0

------- while i < n and j < len(pattern):
----------- if s[i] != pattern[j]:
--------------- return ord(s[i]) - ord(pattern[j])
----------- i += 1
----------- j += 1

------- if j == len(pattern):
----------- return 0
------- return -1

--- def lower_bound(self, pattern):
------- l, r = 0, self.n - 1
------- res = self.n

------- while l <= r:
----------- m = (l + r) // 2
----------- if self._cmp(self.suffix[m], pattern) >= 0:
--------------- res = m
--------------- r = m - 1
----------- else:
--------------- l = m + 1

------- return res

--- def upper_bound(self, pattern):
------- l, r = 0, self.n - 1
------- res = self.n

------- while l <= r:
----------- m = (l + r) // 2
----------- if self._cmp(self.suffix[m], pattern) > 0:
--------------- res = m
--------------- r = m - 1
----------- else:
--------------- l = m + 1

------- return res

--- def count_occurrences(self, pattern):
------- L = self.lower_bound(pattern)
------- R = self.upper_bound(pattern)
------- return max(0, R - L)
```

9.0.8. *Booth Min String Rotation.*

```
def booth(s: str) -> int:
--- s = s + s
--- n = len(s)

--- f = [-1] * n
--- k = 0

--- for j in range(1, n):
------- i = f[j - k - 1]

------- while i != -1 and s[j] != s[k + i + 1]:
----------- if s[j] < s[k + i + 1]:
--------------- k = j - i - 1
----------- i = f[i]

------- if i == -1 and s[j] != s[k]:
----------- if s[j] < s[k]:
--------------- k = j
----------- f[j - k] = -1
------- else:
----------- f[j - k] = i + 1

--- return k
```

10. Other Algorithms

10.0.1. *Longest Increasing Subsequence.*

```
def lis(arr):
--- if not arr:
------- return []

--- n = len(arr)
--- seq = []
--- back = [0] * n

--- for i in range(n):
------- lo, hi = 0, len(seq)
------- while lo < hi:
----------- mid = (lo + hi) // 2
----------- if arr[seq[mid]] < arr[i]:
--------------- lo = mid + 1
----------- else:
--------------- hi = mid

------- res = lo
------- if res < len(seq):
----------- seq[res] = i
------- else:
----------- seq.append(i)
------- back[i] = -1 if res == 0 else seq[res - 1]

--- ans = []
--- at = seq[-1]
--- while at != -1:
------- ans.append(at)
------- at = back[at]

--- return ans[::-1]
```

10.0.2. *Cycle Find.*

```
def find_cycle(x0, f):
--- t = f(x0)
--- h = f(f(x0))

--- while t != h:
------- t = f(t)
------- h = f(f(h))

--- mu = 0
--- h = x0
--- while t != h:
------- t = f(t)
------- h = f(h)
------- mu += 1

--- lam = 1
--- h = f(t)
--- while t != h:
------- h = f(h)
------- lam += 1

--- return mu, lam
```

10.0.3. *2Sat.*

```
class TwoSAT:
--- def __init__(self, n):
------- self.n = n
------- self.N = 2 * n
------- self.adj = [[] for _ in range(self.N)]

--- def var(self, x):
------- return x << 1

--- def neg(self, x):
------- return x ^ 1

--- def add_imp(self, x, y):
------- self.adj[x].append(y)

--- def add_or(self, x, y):
------- self.add_imp(self.neg(x), y)
------- self.add_imp(self.neg(y), x)

--- def sat(self):
------- n = self.N
------- adj = self.adj

------- index = 0
------- stack = []
------- onstack = [False] * n
------- ids = [-1] * n
------- low = [0] * n
------- comp = [-1] * n
------- self.time = 0
------- self.comp_id = 0

------- def dfs(u):
----------- nonlocal index

----------- ids[u] = low[u] = index
----------- index += 1
----------- stack.append(u)
----------- onstack[u] = True

----------- for v in adj[u]:
--------------- if ids[v] == -1:
------------------- dfs(v)
------------------- low[u] = min(low[u], low[v])
--------------- elif onstack[v]:
------------------- low[u] = min(low[u], ids[v])

----------- if low[u] == ids[u]:
--------------- while True:
------------------- v = stack.pop()
------------------- onstack[v] = False
------------------- comp[v] = self.comp_id
------------------- if v == u:
----------------------- break
--------------- self.comp_id += 1

------- for i in range(n):
----------- if ids[i] == -1:
--------------- dfs(i)

------- for i in range(0, n, 2):
----------- if comp[i] == comp[i ^ 1]:
--------------- return False

------- return True
```

10.0.4. *Dpll.*

```
class SAT:
--- def __init__(self):
------- self.clauses = []
------- self.n = 0

--- def var(self):
------- self.n += 1
------- return self.n

--- def clause(self, c):
------- self.clauses.append(c)
```

```
--- def clause_sat(self, clause, assign):
------- unknown = False

------- for x in clause:
----------- v = abs(x)
----------- val = assign[v]

----------- if x > 0 and val == 1:
--------------- return True
----------- if x < 0 and val == 0:
--------------- return True

----------- if val == -1:
--------------- unknown = True

------- return False if not unknown else None

--- def propagate(self, assign):
------- changed = True

------- while changed:
----------- changed = False

----------- for clause in self.clauses:
--------------- sat = False
--------------- unassigned = []

--------------- for x in clause:
------------------- v = abs(x)

------------------- if x > 0 and assign[v] == 1:
----------------------- sat = True
----------------------- break
------------------- if x < 0 and assign[v] == 0:
----------------------- sat = True
----------------------- break

------------------- if assign[v] == -1:
----------------------- unassigned.append(x)

--------------- if sat:
------------------- continue

--------------- if len(unassigned) == 0:
------------------- return False  # contradiction

--------------- if len(unassigned) == 1:
------------------- x = unassigned[0]
------------------- v = abs(x)
------------------- val = 1 if x > 0 else 0

------------------- if assign[v] != -1 and assign[v] != val:
----------------------- return False

------------------- if assign[v] == -1:
----------------------- assign[v] = val
----------------------- changed = True

------- return True

--- def solve_dfs(self, assign, idx=1):
------- if idx > self.n:
----------- return True

------- if assign[idx] != -1:
----------- return self.solve_dfs(assign, idx + 1)

------- for val in [0, 1]:
----------- new_assign = assign[:]
----------- new_assign[idx] = val

----------- if self.propagate(new_assign):
--------------- if self.solve_dfs(new_assign, idx + 1):
------------------- return True

------- return False

--- def solve(self):
------- assign = [-1] * (self.n + 1)

------- if not self.propagate(assign):
----------- return False

------- return self.solve_dfs(assign)

--- def get_value(self, x):
------- return self.assign[x] == 1
```

10.0.5. *Stable Marriage.*

```
from collections import deque

def stable_marriage(n, m_pref, w_pref):
--- w_rank = [[0] * n for _ in range(n)]

--- for w in range(n):
------- for i, man in enumerate(w_pref[w]):
----------- w_rank[w][man] = i

--- q = deque(range(n))

--- at = [0] * n

--- eng = [-1] * n

--- res = [-1] * n

--- while q:
------- m = q.popleft()

------- while at[m] < n:
----------- w = m_pref[m][at[m]]
----------- at[m] += 1

----------- if eng[w] == -1:
--------------- eng[w] = m
--------------- res[m] = w
--------------- break

----------- if w_rank[w][m] < w_rank[w][eng[w]]:
--------------- q.append(eng[w])
--------------- eng[w] = m
--------------- res[m] = w
--------------- break

--- return res
```

10.0.6. *Dates.*

```
def c_div(a, b):
--- return int(a / b)

def date_to_int(y, m, d):
--- k = c_div(m - 14, 12)

--- return (
------- 1461 * (y + 4800 + k) // 4
------- + 367 * (m - 2 - k * 12) // 12
------- - 3 * ((y + 4900 + k) // 100) // 4
------- + d - 32075
--- )

def int_to_day(jd):
--- return jd % 7

def int_to_date(jd):
--- x = jd + 68569
--- n = 4 * x // 146097
--- x -= (146097 * n + 3) // 4

--- i = (4000 * (x + 1)) // 1461001
--- x -= 1461 * i // 4 - 31

--- j = 80 * x // 2447
--- d = x - 2447 * j // 80

--- x = j // 11
--- m = j + 2 - 12 * x
--- y = 100 * (n - 49) + i + x

--- return y, m, d
```

10.0.7. *Simulated Annealing.*

```
import random
import math
import time

def simulated_annealing(n, seconds):
--- rng = random.Random()

--- sol = list(range(1, n + 1))
--- rng.shuffle(sol)

--- def score(arr):
------- return sum(abs(arr[i] - arr[i - 1]) for i in range(1, n))

--- cur = score(sol)

--- T0 = 100.0
--- T1 = 0.001
--- start = time.time()
--- iters = 0

--- while True:
------- if iters % 16 == 0:
----------- progress = (time.time() - start) / seconds
----------- if progress >= 1.0:
--------------- break
----------- temp = T0 * (T1 / T0) ** progress

------- a = rng.randint(0, n - 2)

------- x, y = sol[a], sol[a + 1]

------- delta = 0

------- if a > 0:
----------- delta += abs(sol[a - 1] - y) - abs(sol[a - 1] - x)

------- if a + 2 < n:
----------- delta += abs(x - sol[a + 2]) - abs(y - sol[a + 2])

------- accept = delta <= 0 or rng.random() < math.exp(-delta / temp)

------- if accept:
----------- sol[a], sol[a + 1] = sol[a + 1], sol[a]
----------- cur += delta

------- iters += 1

--- return cur, sol
```

10.0.8. *Fast Input Reading.*

```
import sys ------------------------------------------------------------
 ----------------------------------------------------------------------
data = sys.stdin.buffer.read().split() -------------------------------
it = iter(data) -------------------------------------------------------
 ----------------------------------------------------------------------
def next_int(): -------------------------------------------------------
--- return int(next(it)) ------------------------------------------------
```

10.0.9. *Bit Hacks.*

```
def snoob(x): ---------------------------------------------------------
--- y = x & -x ------------------------------------------------------
--- z = x + y -------------------------------------------------------
--- return z | (((x ^ z) // y) >> 2) --------------------------------
```